Checkpoint B - full solutions

Full solutions

Use these solutions only after completing the 75-minute attempt. Review each method, result and common mistake, then record the first cause of any lost mark.

Question 1 — quadratic forms and convexity

Part A — CPB-Q01-PA (M8) Method and conditions. A quadratic form depends only on the symmetric part of its matrix. Apply a valid definiteness test to that symmetric matrix. Then use the connection between concavity and convex upper contour sets.

The symmetric part is

$$\mathbf{S} = \frac{\mathbf{A} + \mathbf{A}^{\mathsf{T}}}{2} = \begin{bmatrix} -2 & 1 \\ 1 & -1 \end{bmatrix}.$$

Indeed,

$$q(\mathbf{x}) = \mathbf{x}^{\mathsf{T}}\mathbf{S}\mathbf{x} = -2x_1^2 + 2x_1x_2 - x_2^2.$$

The leading principal minors of $\mathbf{S}$ are

$$\Delta_1 = -2 < 0, \qquad \Delta_2 = \det(\mathbf{S}) = 2 - 1 = 1 > 0.$$

The alternating signs show that $\mathbf{S}$ is negative definite. Therefore q is strictly concave on $\mathbb{R}^2$.

To check the set directly, take $\mathbf{u}, \mathbf{v} \in C$ and $t \in [0, 1]$. Concavity gives

$$q(t\mathbf{u} + (1-t)\mathbf{v}) \geq tq(\mathbf{u}) + (1-t)q(\mathbf{v}) \geq -t - (1-t) = -1.$$

Thus $t\mathbf{u} + (1-t)\mathbf{v} \in C$.

Final result. The quadratic form is negative definite, and $C = \{\mathbf{x} : q(\mathbf{x}) \geq -1\}$ is convex.

Marks (4). Correct symmetric part: 1. Valid definiteness calculation: 1. Negative-definite and concavity conclusion: 1. Valid convexity justification for the superlevel set: 1.

Common mistakes. Applying a symmetry-based test directly to $\mathbf{A}$; using the positive-definite sign pattern; claiming that every level set of a concave function is convex rather than identifying a superlevel set.

Question 2 — unconstrained optimisation

Part A — CPB-Q02-PA (M9) Method and conditions. Solve the interior stationary equations. Use the Hessian first for local classification and then on the whole convex domain for a global strict-concavity conclusion.

The column gradient is

$$\nabla f(x, y) = \begin{bmatrix} 6 - 4x \\ 8 - 2y \end{bmatrix}.$$

Hence the unique stationary point is

$$(x^*, y^*) = \left(\frac{3}{2}, 4\right).$$

The Hessian is constant:

$$\mathbf{H}_f(x,y) = \begin{bmatrix} -4 & 0 \\ 0 & -2 \end{bmatrix}.$$

It is negative definite. The stationary point is therefore a strict local maximum. Because the Hessian is negative definite everywhere, f is strictly concave on the convex domain $\mathbb{R}^2$. Consequently the stationary point is the unique global maximiser.

Its value is

$$f\left(\frac{3}{2}, 4\right) = 9 + 32 - \frac{9}{2} - 16 = \frac{41}{2}.$$

Final result. The unique global maximiser is $(3/2, 4)$, and the global maximum value is $41/2$.

Marks (5). Correct gradient or stationary equations: 1. Correct stationary point: 1. Negative-definite Hessian and local classification: 1. Valid strict-concavity global argument: 1. Correct optimum value: 1.

Common mistakes. Treating stationarity alone as sufficient; checking the Hessian only at the candidate and asserting a global result; making an arithmetic error when evaluating the objective.

Question 3 — equality-constrained optimisation and the envelope theorem

Part A — CPB-Q03-PA (M10) Method and conditions. Use the stated constraint encoding. The equality constraint has non-zero gradient $[-2, -1]^\mathsf{T}$, so its regularity condition holds. Solve the first-order system on the positive domain and use a restriction-to-the-feasible-line argument for uniqueness.

The Lagrangian is

$$\mathcal{L}(x, y, \mu; a) = \sqrt{xy} + \mu(a - 2x - y).$$

The first-order and feasibility conditions are

$$\frac{1}{2}\sqrt{\frac{y}{x}} - 2\mu = 0, \qquad \frac{1}{2}\sqrt{\frac{x}{y}} - \mu = 0, \qquad a - 2x - y = 0.$$

The first two equations imply

$$\mu = \frac{1}{4}\sqrt{\frac{y}{x}} = \frac{1}{2}\sqrt{\frac{x}{y}},$$

so $y = 2x$. Feasibility then gives

$$x^*(a) = \frac{a}{4}, \qquad y^*(a) = \frac{a}{2}, \qquad \mu^*(a) = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}.$$

These values are positive for every $a > 0$.

For a sufficiency and uniqueness check, substitute $y = a - 2x$. On $0 < x < a/2$, maximising $\sqrt{x(a - 2x)}$ is equivalent to maximising

$$ax - 2x^2,$$

which is strictly concave and has the unique maximiser $x = a/4$. Thus the displayed feasible point is the unique global maximiser.

At $a = 12$,

$$(x^*, y^*, \mu^*) = \left(3, 6, \frac{1}{2\sqrt{2}}\right).$$

Final result.

$$x^*(a) = \frac{a}{4}, \qquad y^*(a) = \frac{a}{2}, \qquad \mu^*(a) = \frac{1}{2\sqrt{2}}.$$

Marks (5). Correct Lagrangian and first-order system: 1. Derives $y = 2x$: 1. Derives $x^*(a)$ and $y^*(a)$: 1. Correct multiplier and values at $a = 12$: 1. Valid sufficiency and uniqueness argument: 1.

Common mistakes. Losing the factor 2 from the constraint derivative; reversing the ratio in a square-root derivative; calling $\sqrt{xy}$ strictly concave on the whole positive orthant without a valid restriction argument; omitting the multiplier.

Part B — CPB-Q03-PB (M10) Method and conditions. Evaluate the objective at the optimiser. Under the encoding $a - 2x - y = 0$, the envelope derivative equals the multiplier because the derivative of the constraint function with respect to a is one.

The value function is

$$V(a) = \sqrt{\frac{a}{4}\frac{a}{2}} = \frac{a}{2\sqrt{2}} = \frac{\sqrt{2}}{4}a, \qquad a > 0.$$

The envelope theorem gives

$$V'(a) = \left.\frac{\partial \mathcal{L}}{\partial a}\right|_{*} = \mu^*(a) = \frac{1}{2\sqrt{2}}.$$

Direct differentiation of $V(a)$ gives the same derivative.

Final result and interpretation. $V'(a) = 1/(2\sqrt{2}) > 0$. Under the stated encoding, the positive multiplier is the marginal increase in maximised utility from one additional unit of the resource parameter.

Marks (3). Correct value function: 1. Correct envelope derivative and direct verification: 1. Correct sign interpretation under the stated encoding: 1.

Common mistakes. Omitting the constraint encoding when interpreting the multiplier; differentiating the optimisers unnecessarily and losing terms; reporting the multiplier with the opposite sign.

Question 4 — a Kuhn–Tucker corner

Part A — CPB-Q04-PA (M11) Method and conditions. Treat all three inequalities as explicit $g_j \geq 0$ constraints. Write stationarity, primal feasibility, dual feasibility and complementary slackness before identifying the active branch.

Let λ_x, λ_y and μ be the multipliers on $x \geq 0$, $y \geq 0$ and $1 - x - y \geq 0$, respectively. The Lagrangian is

$$\mathcal{L} = 4x + y - x^2 - y^2 + \lambda_x x + \lambda_y y + \mu(1 - x - y).$$

The complete conditions are

$$4 - 2x + \lambda_x - \mu = 0, \qquad 1 - 2y + \lambda_y - \mu = 0,$$

$$x \geq 0, \qquad y \geq 0, \qquad 1 - x - y \geq 0,$$

$$\lambda_x \geq 0, \qquad \lambda_y \geq 0, \qquad \mu \geq 0,$$

$$\lambda_x x = 0, \qquad \lambda_y y = 0, \qquad \mu(1 - x - y) = 0.$$

The unconstrained maximiser $(2, 1/2)$ is infeasible. The resource constraint cannot be slack at an optimum because $4 - 2x > 0$ throughout the feasible range $0 \leq x \leq 1$. If both variables were positive while the resource constraint bound, then $\lambda_x = \lambda_y = 0$. Stationarity and $x + y = 1$ would imply $x = 5/4$, which is infeasible. Hence the binding solution is a corner with $y = 0$, so feasibility gives $x = 1$.

At $(x, y) = (1, 0)$, complementary slackness and stationarity give

$$\lambda_x = 0, \qquad \mu = 2, \qquad \lambda_y = 1.$$

Thus $x \geq 0$ is slack, while $y \geq 0$ and $1 - x - y \geq 0$ are active. The Hessian of the objective is $-2\mathbf{I}_2$, which is negative definite, and the feasible triangle is convex. The feasible Kuhn–Tucker point is therefore the unique global maximiser.

Final result.

$$(x^*, y^*) = (1, 0), \qquad (\lambda_x^*, \lambda_y^*, \mu^*) = (0, 1, 2).$$

Marks (6). Correct Lagrangian: 1. Complete stationarity and feasibility conditions: 2. Correct corner and active/slack classification: 1. Correct multipliers: 1. Valid unique-global sufficiency argument: 1.

Common mistakes. Writing a $g_j \geq 0$ constraint with a negative Lagrangian sign; omitting the lower-bound multiplier for y; calling $x \geq 0$ active at $x = 1$; assuming that an active constraint must have a strictly positive multiplier in every problem.

Question 5 — a joint density and conditional expectation

Part A — CPB-Q05-PA (S5) Method and conditions. Integrate the joint density over the other variable, retain the rectangular support, and divide by the positive marginal density. Independence requires factorisation on the joint support.

The density is normalised because

$$\int_0^1 \int_0^1 (x + y)\, \mathrm{d}x\, \mathrm{d}y = 1.$$

For $0 < y < 1$,

$$f_Y(y) = \int_0^1 (x + y)\, \mathrm{d}x = \frac{1}{2} + y,$$

and $f_Y(y) = 0$ otherwise. Therefore, for $0 < y < 1$,

$$f_{X|Y}(x \mid y) = \frac{x+y}{y+1/2}, \qquad 0 < x < 1,$$

and the conditional density is zero outside that conditional support.

The conditional expectation is

$$\begin{aligned}\mathbb{E}[X \mid Y = y] &= \frac{\int_0^1 x(x+y)\,\mathrm{d}x}{y+1/2} \\ &= \frac{1/3 + y/2}{y+1/2} = \frac{2+3y}{3+6y}, \qquad 0 < y < 1.\end{aligned}$$

Also $f_X(x) = x + 1/2$ for $0 < x < 1$. In general,

$$x + y \neq (x+1/2)(y+1/2),$$

so $f_{X,Y} \neq f_X f_Y$. Equivalently, the conditional density depends on y.

Final result. The marginal and conditional densities are as displayed, $\mathbb{E}[X \mid Y = y] = (2+3y)/(3+6y)$, and X and Y are not independent.

Marks (5). Correct marginal density and support: 1. Correct conditional density and support: 1. Correct conditional-expectation setup: 1. Correct conditional-expectation value: 1. Valid non-independence justification: 1.

Common mistakes. Integrating over the wrong variable; omitting the conditional support; failing to divide by $f_Y(y)$; inferring independence from the rectangular support alone.

Question 6 — exact and approximate sampling distributions

Part A — CPB-Q06-PA (S6) Method and conditions. Independence makes the sum of the Poisson variables exactly Poisson. For the sample mean, use the i.i.d. mean and variance formulas, then apply the central limit theorem as requested.

Poisson additivity gives

$$T \sim \text{Poisson}(32 \cdot 2) = \text{Poisson}(64)$$

exactly. Moreover,

$$\mathbb{E}[\bar{X}] = 2, \qquad \text{Var}(\bar{X}) = \frac{2}{32} = \frac{1}{16},$$

so the standard deviation of $\bar{X}$ is $1/4$. Without a continuity correction,

$$\begin{aligned}\mathbb{P}(\bar{X} > 2.5) &\approx \mathbb{P}\left(Z > \frac{2.5-2}{1/4}\right) \\ &= 1 - \Phi(2) = 1 - 0.9772 = 0.0228.\end{aligned}$$

The distribution of T is exact because independent Poisson rates add. The finite-sample distribution of $\bar{X} = T/32$ is discrete rather than normal, so the final normal-tail calculation is a central-limit approximation.

Final result.

$$T \sim \text{Poisson}(64), \qquad \mathbb{E}[\bar{X}] = 2, \qquad \text{Var}(\bar{X}) = \frac{1}{16}, \qquad \mathbb{P}(\bar{X} > 2.5) \approx 0.0228.$$

Marks (5). Exact distribution of T: 1. Correct mean and variance of $\bar{X}$: 1. Correct standardisation: 1. Correct approximate probability: 1. Correct exact-versus-approximate explanation: 1.

Common mistakes. Using a Poisson variance of 2^2; dividing the sample-mean variance by 32^2 after already using a single-observation variance; standardising with variance rather than standard deviation; calling the final normal probability exact.

Question 7 — method of moments and maximum likelihood

Part A — CPB-Q07-PA (S7) Method and conditions. Match the population mean to the sample mean for method of moments. For maximum likelihood, state the parameter space and treat the interior and boundary samples separately. Use the law of large numbers for consistency.

Let $S := \sum_{i=1}^{n} x_i$. The moment equation is

$$\bar{X} = \frac{1}{p},$$

so

$$\hat{p}_{\text{MM}} = \frac{1}{\bar{X}}.$$

The likelihood is

$$L(p; \mathbf{x}) = p^n (1-p)^{S-n}, \qquad 0 < p \leq 1.$$

If $S > n$, then $0 < p < 1$ at an interior maximum and

$$\ell'(p) = \frac{n}{p} - \frac{S-n}{1-p}.$$

Solving $\ell'(p) = 0$ gives

$$\hat{p}_{\text{MLE}} = \frac{n}{S} = \frac{1}{\bar{X}}.$$

Moreover,

$$\ell''(p) = -\frac{n}{p^2} - \frac{S-n}{(1-p)^2} < 0,$$

so this is the unique maximum. If $S = n$, every observation is one and the likelihood is p^n, which is uniquely maximised at the permitted boundary $p = 1$. The same formula $n/S = 1/\bar{X}$ therefore covers both cases.

The method-of-moments and maximum-likelihood estimators coincide in this model. For $(1, 2, 1, 4)$, $n = 4$, $S = 8$ and

$$\hat{p} = \frac{4}{8} = \frac{1}{2}.$$

For $0 < p < 1$, the law of large numbers gives $\bar{X} \to 1/p$ in probability. Because inversion is continuous at the positive value $1/p$,

$$\frac{1}{\bar{X}} \longrightarrow p$$

in probability. At $p = 1$, all observations equal one almost surely and the estimator equals one exactly. Thus the estimator is consistent throughout $0 < p \leq 1$.

Final result.

$$\widehat{p}_{\mathrm{MM}} = \widehat{p}_{\mathrm{MLE}} = \frac{1}{\bar{X}} = \frac{n}{\sum_{i=1}^{n} X_i},$$

with observed estimate $1/2$ for the supplied sample.

Marks (7). Correct method-of-moments estimator: 1. Correct likelihood and parameter space: 1. Correct interior score solution: 2. Correct boundary sample treatment: 1. Correct relationship and observed estimate: 1. Valid consistency argument, including the parameter boundary: 1.

Common mistakes. Using the geometric failures-before-success parameterisation; omitting the parameter boundary $p = 1$; treating the score equation as sufficient without a maximum check; asserting consistency from one observed estimate rather than a repeated-sampling argument.